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# FSL

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Fundamentals of statistical learning

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# 1

## Probability

### 1.1 Why probability?

Observed data are the results of processes whose behavior we may not fully know. To reason about them, we describe the process with a probabilistic model. Probability lets us work forward from a model to possible outcomes; statistical inference uses observed outcomes to learn about the model.

#### Example 1: A coin toss

Let  $\theta = P(\text{heads})$  be the probability that a coin lands heads. If  $\theta$  is known, we can ask how likely it is to observe three heads in five tosses. If instead we observe three heads in five tosses while  $\theta$  is unknown, we can use those data to learn about  $\theta$ .

### 1.2 Sample spaces and events

Before assigning probabilities, we must specify what the experiment can produce. We start with a finite set of possible results.

#### Definition 1.1: Outcomes and events

A random experiment is a process whose result is not known in advance. Each possible result is an *outcome*  $\omega$ . The set of all possible outcomes is the *sample space*  $\Omega$ . An *event* is a subset  $A \subseteq \Omega$ ; it occurs when the observed outcome belongs to  $A$ .

#### Example 2: Rolling a die

For one roll of a six-sided die, the sample space is  $\Omega = \{1, 2, 3, 4, 5, 6\}$ . The event “the result is even” is  $A = \{2, 4, 6\}$ . The whole sample space  $\Omega$  is the certain event, while  $\emptyset$  is the impossible event.

### 1.2.1 Operations on events

Events are subsets of the same sample space  $\Omega$ , so set operations let us describe new events in terms of  $A$  and  $B$ :

- **Union:**  $A \cup B := \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$ . This event occurs when at least one of  $A$  and  $B$  occurs; “or” includes the case in which both occur. In particular,  $A \cup \emptyset = A$  and  $A \cup \Omega = \Omega$ .
- **Intersection:**  $A \cap B := \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$ . This event occurs when both  $A$  and  $B$  occur. In particular,  $A \cap \Omega = A$  and  $A \cap \emptyset = \emptyset$ .
- **Complement:**  $A^c := \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$ . This event occurs when  $A$  does not occur. In particular,  $A \cup A^c = \Omega$  and  $A \cap A^c = \emptyset$ .

For the die in Example 2, let  $B = \{4, 5, 6\}$  be the event “the result is at least four”. Then  $A \cup B = \{2, 4, 5, 6\}$ ,  $A \cap B = \{4, 6\}$ , and  $A^c = \{1, 3, 5\}$ .

Two events are *disjoint* when  $A \cap B = \emptyset$ , so they cannot occur together. If  $A \subseteq B$ , every outcome in  $A$  is also in  $B$ : whenever  $A$  occurs,  $B$  occurs.

#### Proposition 1.1: De Morgan’s laws

For any events  $A$  and  $B$ ,

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

More generally, for events  $A_1, \dots, A_n$ ,

$$\left( \bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c, \quad \left( \bigcap_{i=1}^n A_i \right)^c = \bigcup_{i=1}^n A_i^c.$$

The first identity says that “none occurs” means that every event fails. The second says that “not all occur” means that at least one event fails. We will use the second identity in the proof of Bonferroni’s inequality.

### 1.2.2 Partitions and event decomposition

A useful way to study an event is to split it into disjoint pieces. For any events  $A$  and  $B$ , exactly one of  $B$  and  $B^c$  occurs, so

$$A = A \cap \Omega = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c).$$

The two events on the right are disjoint. For example, a car can fail an emissions test either while it pollutes or while it does not.

More generally, a *partition*  $B_1, \dots, B_n$  of  $\Omega$  consists of pairwise disjoint events whose union is  $\Omega$ . Every event  $A$  can then be written as

$$A = \bigcup_{i=1}^n (A \cap B_i), \quad (A \cap B_i) \cap (A \cap B_j) = \emptyset \quad (i \neq j).$$

This decomposition will let us add the probabilities of different ways in which the same event can occur.

## 1.3 Probability

There are several ways to interpret a probability. Under a *symmetry* assumption, equally likely outcomes give  $P(A) = |A|/|\Omega|$ . A *frequency* interpretation concerns the long-run proportion of repetitions in which  $A$  occurs. A *subjective* interpretation expresses a degree of belief about  $A$ . The axioms below state the mathematical rules that a probability assignment must satisfy, regardless of its interpretation.

### Example 3: Two dice

If the 36 ordered results of rolling two dice are equally likely, they form the sample space  $\Omega = \{1, \dots, 6\}^2$ . The sum is eight for  $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$ , so the probability of that event is  $5/36$ . Counting outcomes this way is valid because the ordered pairs are equally likely.

### Definition 1.2: Probability on a finite sample space

A probability measure  $P$  assigns a number  $P(A)$  to every event  $A \subseteq \Omega$  such that:

1.  $0 \leq P(A) \leq 1$  for every event  $A$ ;
2.  $P(\Omega) = 1$ ;
3. if  $A \cap B = \emptyset$ , then  $P(A \cup B) = P(A) + P(B)$ .

The last property is called *finite additivity*.

Because  $A$  and  $A^c$  are disjoint and their union is  $\Omega$ , these axioms give  $P(A^c) = 1 - P(A)$ . In particular,  $P(\emptyset) = 0$ .

They also imply *monotonicity*: if  $A \subseteq B$ , then  $B = A \cup (B \cap A^c)$  is a disjoint union, so  $P(B) = P(A) + P(B \cap A^c) \geq P(A)$ .

### Example 4: A fair die

If the six outcomes of a die roll are equally likely, each has probability  $1/6$ . The event  $A = \{2, 4, 6\}$  therefore has probability  $P(A) = 3/6 = 1/2$ .

For an infinite sample space, finite additivity is replaced by *countable additivity*: for pairwise disjoint events  $A_1, A_2, \dots$ ,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

In that setting, a probability measure is defined on a suitable family of events rather than necessarily on every subset of  $\Omega$ . On a finite sample space, the finite formulation above is enough.

### 1.3.1 Inclusion–exclusion

Finite additivity directly handles disjoint events. If  $A$  and  $B$  overlap, adding  $P(A)$  and  $P(B)$  counts their intersection twice.

**Proposition 1.2: Inclusion–exclusion for two events**

For any events  $A$  and  $B$ ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

*Proof.* Split  $A \cup B$  into the disjoint events  $A$  and  $B \cap A^c$ :

$$P(A \cup B) = P(A) + P(B \cap A^c).$$

The event  $B$  is itself the disjoint union of  $A \cap B$  and  $A^c \cap B$ . Thus  $P(B \cap A^c) = P(B) - P(A \cap B)$ , which gives the formula.  $\square$

**Example 5: Overlapping die events**

Let  $A = \{2, 4, 6\}$  and  $B = \{4, 5, 6\}$ . Then  $P(A) = P(B) = 1/2$  and  $P(A \cap B) = 1/3$ . Therefore

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \frac{2}{3},$$

in agreement with  $A \cup B = \{2, 4, 5, 6\}$ .

Applying the two-event formula twice gives the three-event version:

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C). \end{aligned}$$

The final term restores outcomes in all three events: they were added three times and subtracted three times.

For  $n$  events, the same pattern continues:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{\emptyset \neq I \subseteq \{1, \dots, n\}} (-1)^{|I|+1} P\left(\bigcap_{i \in I} A_i\right).$$

To see why, consider an outcome contained in exactly  $k$  of the events. It is counted  $\binom{k}{j}$  times among the intersections of size  $j$ , so its total coefficient on the right is  $\sum_{j=1}^k (-1)^{j+1} \binom{k}{j} = 1$ . An outcome in none of the events contributes zero to both sides.

### 1.3.2 Probability bounds

Sometimes the probabilities of intersections are unknown. We can still bound the probability of a union.

**Proposition 1.3: Union bound (Boole's inequality)**

For events  $A_1, \dots, A_n$ ,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

*Proof.* We prove the bound by induction on the number of events.

**Base case** ( $n = 1$ ). The union contains only  $A_1$ , so the inequality is an equality:  $P(A_1) = \sum_{i=1}^1 P(A_i)$ .

**Inductive step.** Assume the bound holds for  $n$  events and set  $U_n = \bigcup_{i=1}^n A_i$ . The union of  $n + 1$  events is  $U_n \cup A_{n+1}$ . Applying inclusion–exclusion gives

$$\begin{aligned} P\left(\bigcup_{i=1}^{n+1} A_i\right) &= P(U_n) + P(A_{n+1}) - P(U_n \cap A_{n+1}) \\ &\leq P(U_n) + P(A_{n+1}) \\ &\leq \sum_{i=1}^n P(A_i) + P(A_{n+1}) = \sum_{i=1}^{n+1} P(A_i). \end{aligned}$$

The first inequality uses  $P(U_n \cap A_{n+1}) \geq 0$ ; the second uses the induction hypothesis. Thus the bound holds for every  $n \geq 1$ .  $\square$

For a countable sequence, the same inequality follows by applying the finite bound to the first  $n$  events and taking the limit.

**Proposition 1.4: Bonferroni's inequality**

For events  $A_1, \dots, A_n$ ,

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n - 1).$$

*Proof.* The second  $n$ -event form of De Morgan's law turns the complement of an intersection into a union of complements. Then the union bound gives

$$\begin{aligned} P\left(\bigcap_{i=1}^n A_i\right) &= 1 - P\left(\bigcup_{i=1}^n A_i^c\right) \\ &\geq 1 - \sum_{i=1}^n P(A_i^c) \\ &= 1 - \sum_{i=1}^n (1 - P(A_i)) = \sum_{i=1}^n P(A_i) - (n - 1). \end{aligned}$$

$\square$

### 1.3.3 Application: a randomized identity test

Suppose  $F$  and  $G$  are polynomials of degree at most  $d$ . We want to test whether they are identical without comparing all their coefficients. Choose a nonempty finite set  $S$  of possible evaluation points.

#### Algorithm 1.1: Randomized polynomial identity test

Given  $F$ ,  $G$ , and  $S$ , report whether the polynomials differ or may be identical.

```

function IDENTITYTEST( $F, G, S$ )
  Draw  $R$  uniformly at random from  $S$ 
  if  $F(R) \neq G(R)$  then
    return DIFFERENT
  else
    return POSSIBLY EQUAL
  end if
end function

```

If  $F = G$ , the test never reports a difference. If  $F \neq G$ , it fails to detect the difference only when  $R$  belongs to the event

$$E = \{r \in S : F(r) = G(r)\}.$$

Indeed,  $E$  contains the roots in  $S$  of the nonzero polynomial  $H = F - G$ , which has at most  $d$  roots. Because  $R$  is uniform on  $S$ ,

$$P(R \in E) = \frac{|E|}{|S|} \leq \frac{d}{|S|}.$$

For  $S = \{1, \dots, 100d\}$  with  $d \geq 1$ , the probability of missing a difference is at most  $1/100$ . The possible error is *one-sided*: “different” is certain, while “possibly equal” is not a proof of identity.

## 1.4 Independent events

### Definition 1.3: Independent events

Two events  $A$  and  $B$  are independent if

$$P(A \cap B) = P(A)P(B).$$

When  $P(B) > 0$ , this is equivalent to  $P(A \mid B) = P(A)$ : knowing that  $B$  occurred does not change the probability of  $A$ .

## 1.5 Conditional probability

## 1.6 Bayes' theorem

### 1.6.1 Odds

For an event  $A$  with  $P(A) < 1$ , its *odds in favor* are

$$O(A) := \frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}.$$

Conversely,  $P(A) = O(A)/(1 + O(A))$ . For a fair die, the odds in favor of rolling a six are  $(1/6)/(5/6) = 1/5$ , often stated as one to five. This is different from saying that the probability is  $1/5$ ; the probability is  $1/6$ .

# 2

## Crash course exercises

### 2.1 Day 01

#### Exercise 1: Computer virus

A virus can enter a computer through e-mail or the internet. The probability of entry through e-mail is 0.30, through the internet 0.40, and through both routes 0.15. What is the probability that the virus does not enter at all?

#### Solution.

Let

$E = \{\text{the virus enters through e-mail}\}, \quad I = \{\text{the virus enters through the internet}\}.$

**Step 1: Define the target event.** Let  $N = E^c \cap I^c$  be the event that the virus does not enter. We seek  $P(N)$ .

**Step 2: Apply De Morgan's law.**

$$P(N) = P((E \cup I)^c).$$

**Step 3: Use the complement rule.**

$$P((E \cup I)^c) = 1 - P(E \cup I).$$

**Step 4: Calculate the union by inclusion-exclusion.**

$$P(E \cup I) = P(E) + P(I) - P(E \cap I) = 0.30 + 0.40 - 0.15 = 0.55.$$

**Step 5: Take the complement.**

$$P(N) = 1 - 0.55 = 0.45 = 45\%.$$

**Exercise 2: Two children**

A family has two children, and at least one is a boy. What is the probability that both are boys?

**Solution.**

Assume each child is independently a boy or a girl with probability  $1/2$ . Let

$$B_1 = \{\text{the first child is a boy}\}, \quad B_2 = \{\text{the second child is a boy}\}.$$

**Step 1: Define the events in the question.** Let

$$\begin{aligned} A &= B_1 \cap B_2 = \{\text{both children are boys}\}, \\ C &= B_1 \cup B_2 = \{\text{at least one child is a boy}\}. \end{aligned}$$

We seek  $P(A \mid C)$ .

**Step 2: List the ordered outcomes.** With  $B$  for boy and  $G$  for girl, the equally likely possibilities are

$$\Omega = \{BB, BG, GB, GG\}.$$

**Step 3: Use the information given.** The event  $C$  excludes  $GG$ , leaving

$$C = \{BB, BG, GB\}.$$

**Step 4: Count the favorable outcomes.** We have  $A = \{BB\}$ , so one of the three outcomes in  $C$  is favorable:

$$P(A \mid C) = \frac{1}{3}.$$

**Check using conditional probability.**

Since  $A \subseteq C$ , we have  $A \cap C = A$ . Independence and the complement rule give

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad P(C) = 1 - P(\{GG\}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Therefore

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

If the information had instead been *the first child is a boy*, only  $BB$  and  $BG$  would remain, giving  $1/2$ . The way the information is supplied matters.

## 2.2 Day 02

### Exercise 3: Rolling two dice

Two fair six-sided dice are rolled. Describe the sample space and find the probability that their sum is seven.

#### Solution.

**Step 1: Write the sample space.** The result is an ordered pair  $(i, j)$ , one value for each die:

$$\Omega = \{(i, j) : i, j \in \{1, \dots, 6\}\}, \quad |\Omega| = 6 \cdot 6 = 36.$$

In particular,  $(1, 6)$  and  $(6, 1)$  are different outcomes.

**Step 2: Define the target event.** Let

$$S = \{(i, j) \in \Omega : i + j = 7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

We seek  $P(S)$ , and  $|S| = 6$ .

**Step 3: Apply the equally likely outcomes rule.**

$$P(S) = \frac{|S|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}.$$

### Exercise 4: Five-card poker hands

Assume that all five-card hands from a standard 52-card deck are equally likely. Find the probabilities of a flush (five cards of one suit), four of a kind, and exactly one pair.

#### Solution.

Because order does not matter, all three probabilities have the same denominator:

$$N = \binom{52}{5} = 2\,598\,960.$$

**Flush.**

**Step 1: Choose the suit:** 4 possibilities.

**Step 2: Choose five cards of that suit:**  $\binom{13}{5}$  possibilities.

**Step 3: Count and divide:**

$$P(\text{flush}) = \frac{4 \binom{13}{5}}{N} = \frac{5\,148}{2\,598\,960} \approx 0.001981.$$

Under the exercise's definition, this includes straight flushes.

**Four of a kind.**

**Step 1: Choose the rank:** 13 possibilities. All four cards of that rank are then fixed.

**Step 2: Choose the fifth card:** any of the remaining 48.

**Step 3: Count and divide:**

$$P(\text{four of a kind}) = \frac{13 \cdot 48}{N} = \frac{624}{2\,598\,960} \approx 0.0002401.$$

**Exactly one pair.**

**Step 1: Choose the pair's rank:** 13 possibilities.

**Step 2: Choose its two suits:**  $\binom{4}{2}$  possibilities.

**Step 3: Choose the other three ranks:**  $\binom{12}{3}$  ways. They must be distinct, avoiding another pair or a triple.

**Step 4: Choose their suits:**  $4^3$  possibilities.

**Step 5: Count and divide:**

$$P(\text{exactly one pair}) = \frac{13 \binom{4}{2} \binom{12}{3} 4^3}{N} = \frac{1\,098\,240}{2\,598\,960} \approx 0.42257.$$

## 2.3 Day 04 (24 sept 26)

### Exercise 5: Rome and Florence

The probability that a tourist has visited Rome is 26%, the probability that they have visited Florence is 18%, and the probability that they have visited at least one of the two cities is 38%.

**Question A.** What is the probability that the tourist has visited both Rome and Florence?

**Question B.** What is the probability that the tourist has visited Rome but not Florence?

**Notation.**

Let

$$R = \{\text{the tourist has visited Rome}\}, \quad F = \{\text{the tourist has visited Florence}\}.$$

The given probabilities are

$$P(R) = 0.26, \quad P(F) = 0.18, \quad P(R \cup F) = 0.38.$$

#### Solution to A.

**Step 1: Define the target event.** Let

$$A = R \cap F = \{\text{the tourist has visited both cities}\}.$$

We seek  $P(A)$ .

**Step 2: Apply inclusion–exclusion and substitute.**

$$\begin{aligned} P(A) &= P(R) + P(F) - P(R \cup F) \\ &= 0.26 + 0.18 - 0.38 \\ &= \boxed{0.06 = 6\%}. \end{aligned}$$

#### Solution to B.

**Step 1: Define the target event.** Let

$$B = R \cap F^c = \{\text{the tourist has visited Rome but not Florence}\}.$$

We seek  $P(B)$ .

**Step 2: Subtract the overlap from Rome.**

$$\begin{aligned} P(B) &= P(R) - P(A) \\ &= 0.26 - 0.06 \\ &= \boxed{0.20 = 20\%}. \end{aligned}$$

**Exercise 6: An uncommon disease**

You are diagnosed with an uncommon disease. You know that there is only a 1% chance of getting it. Let

$$D = \{\text{you have the disease}\}, \quad T = \{\text{the test is positive}\}.$$

It is known that the test is imperfect, since

$$P(T \mid D) = 0.98, \quad P(T^c \mid D^c) = 0.95.$$

**Question A.** Given that you test positive, what is the probability that you have the disease?

**Question B.** You take the test again, independently of the first test conditional on whether you have the disease, and it is positive again. Given both positive results, what is the probability that you have the disease?

**Solution to A.**

We seek  $P(D \mid T)$ . The prior probabilities are  $P(D) = 0.01$  and  $P(D^c) = 0.99$ .

**Step 1: Find the false-positive rate.** By the complement rule,

$$P(T \mid D^c) = 1 - P(T^c \mid D^c) = 1 - 0.95 = 0.05.$$

**Step 2: Find the probability of a positive result.** By the law of total probability,

$$\begin{aligned} P(T) &= P(T \mid D)P(D) + P(T \mid D^c)P(D^c) \\ &= 0.98 \cdot 0.01 + 0.05 \cdot 0.99 \\ &= 0.0593. \end{aligned}$$

**Step 3: Apply Bayes' theorem.**

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T)} = \frac{0.0098}{0.0593} \approx 16.53\%.$$

The positive test raises the probability of disease from 1% to about 16.53% under the stated assumptions.

**Solution to B.**

Let

$$\begin{aligned}T_1 &= \{\text{the first test is positive}\}, \\T_2 &= \{\text{the second test is positive}\}.\end{aligned}$$

We seek  $P(D \mid T_1 \cap T_2)$ .

**Step 1: Find the likelihood of two positive results.** Conditional independence gives

$$P(T_1 \cap T_2 \mid D) = 0.98^2 = 0.9604, \quad P(T_1 \cap T_2 \mid D^c) = 0.05^2 = 0.0025.$$

**Step 2: Find the probability of two positive results.** By the law of total probability,

$$\begin{aligned}P(T_1 \cap T_2) &= P(T_1 \cap T_2 \mid D)P(D) + P(T_1 \cap T_2 \mid D^c)P(D^c) \\&= 0.9604 \cdot 0.01 + 0.0025 \cdot 0.99 \\&= 0.012079.\end{aligned}$$

**Step 3: Apply Bayes' theorem.**

$$P(D \mid T_1 \cap T_2) = \frac{0.9604 \cdot 0.01}{0.012079} \approx 79.51\%.$$

The second positive result raises the probability from about 16.53% to about 79.51% under the conditional independence assumption.

# 3

## Random variables

Start of lecture 25 September

### 3.1 From outcomes to data

Imagine picking one student at random from a class. Before the draw, we do not know who we will get; afterwards, we can write down that student's height, weight, and age. Probability describes the draw, while the numbers we write down are the data.

The possible outcomes form  $\Omega$ , and  $\omega \in \Omega$  is the outcome that actually occurs. We write  $(\Omega, \mathcal{F}, P)$  for the probability space:  $\mathcal{F}$  contains the events to which  $P$  assigns probabilities. For a finite  $\Omega$ , as in Chapter 1, every subset can be an event.

#### Definition (informal): Random object

Given a probability space, a *random object* is a map

$$Z : \Omega \longrightarrow \mathcal{T},$$

where  $\mathcal{T}$  is a suitable space of observations for our data-collection process.

#### Definition 3.1: Random object

A *random object* is a measurable\* map from outcomes to possible observations:

$$Z : \Omega \longrightarrow \mathcal{T}, \quad \omega \longmapsto Z(\omega).$$

The set  $\mathcal{T}$  says what kind of thing we record. It could contain numbers, vectors, images, or text messages. Once  $\omega$  is known,  $Z(\omega)$  is the observation we get.

---

\*A rigorous definition of measurability is beyond the scope of this course.

Think of  $Z$  as the rule that turns the outcome into a piece of data. “Measurable” is the technical condition that lets us assign a probability to questions about that data, such as “is the selected student taller than 180 cm?”

## 3.2 Random variables and random vectors

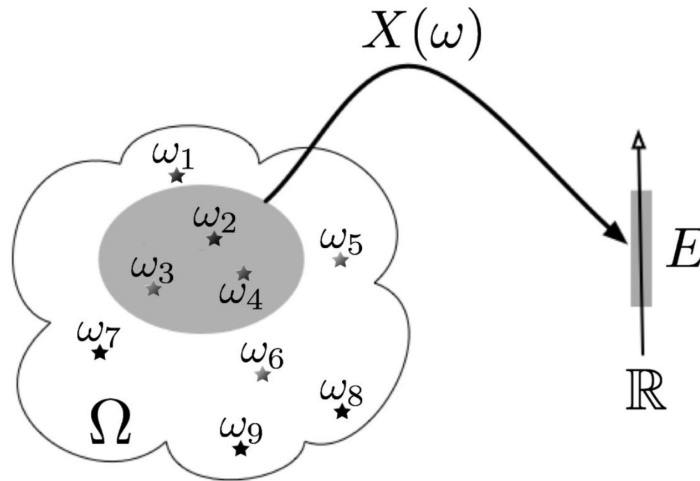
Often the observation is a number or a list of numbers. These are two particularly useful kinds of random objects.

### Definition 3.2: Random variable

A *random variable* is a measurable map

$$X : \Omega \longrightarrow \mathbb{R}.$$

It assigns one real number  $X(\omega)$  to each outcome  $\omega$ .



A map from  $\Omega$  to  $\mathbb{R}$ . Illustration from Brutti’s slides.

### Definition 3.3: Random vector

A  *$p$ -dimensional random vector* is a measurable map

$$\underline{X} : \Omega \longrightarrow \mathbb{R}^p, \quad \underline{X}(\omega) := (X_1(\omega), \dots, X_p(\omega)).$$

Each coordinate  $X_j$  is a real-valued random variable. Thus one outcome gives us  $p$  numbers collected into a vector. These measurements can be related; they do not have to be independent.

**Note: Notation**

In these notes, we denote a random vector by  $\underline{X}$  ( $X$  with a bar underneath):

$$\underline{X} = (X_1, \dots, X_p).$$

**Example 6: One student, one number or three**

Pick a student at random and let  $\omega$  identify the student selected. If  $H$  records height in centimetres, then  $H(\omega)$  is one number. For a student who is 178 cm tall,  $H(\omega) = 178$ .

Now record height  $H$ , weight  $W$  in kilograms, and age  $A$  in years for that *same* student:

$$\underline{X}(\omega) = \begin{pmatrix} H(\omega) \\ W(\omega) \\ A(\omega) \end{pmatrix} = \begin{pmatrix} 178 \\ 72 \\ 21 \end{pmatrix} \in \mathbb{R}^3.$$

The draw gives us one student, not three. We simply take three measurements of that student. Before the draw,  $H$  and  $\underline{X}$  are uncertain; for the selected student, their values are fixed.

In supervised learning, each sampled item gives us an input  $\underline{X}$  (the features) and a target  $Y$  (the label or quantity of interest). Before we see the item, both are uncertain; for an observed item, their values are fixed.

**Example 7: House prices**

Choose a house at random. Use its floor area in square metres, number of rooms, and age in years as features. For a particular house with 85 m<sup>2</sup>, 3 rooms, and an age of 10 years, the data look like this:

$$\underline{X}(\omega) = (85, 3, 10), \quad Y(\omega) = 240,000 \text{ euros.}$$

The three features make a random vector; the sale price is one random variable.

**Example 8: Image labels**

Take a randomly chosen  $64 \times 64$  grayscale photo. Each pixel is one number from 0 (black) to 255 (white). Putting the pixels in a fixed order gives  $\underline{X}(\omega) \in \mathbb{R}^{4096}$ : one vector with 4096 entries for one photo. If we set  $Y(\omega) = 1$  for a cat and  $Y(\omega) = 0$  for a dog, then a cat photo gives one feature vector and the single label 1. The pixels can be related even though they are separate coordinates of  $\underline{X}$ .

### 3.3 Preimages and induced probabilities

To ask a probability question about the values of  $X$ , we first find the outcomes that produce those values.

#### Definition 3.4: Preimage

For a random variable  $X : \Omega \rightarrow \mathbb{R}$  and a set  $E \subseteq \mathbb{R}$ , the *preimage* (or *inverse image*) of  $E$  is

$$X^{-1}(E) := \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega.$$

To find this event, we collect the outcomes whose outputs fall in  $E$ . Whenever  $X^{-1}(E)$  is an event, we define

$$\begin{aligned} P(X \in E) &:= P(X^{-1}(E)) \\ &= P(\{\omega \in \Omega : X(\omega) \in E\}). \end{aligned}$$

In particular, for a single value  $x$ ,

$$P(X = x) := P(X^{-1}(\{x\})).$$

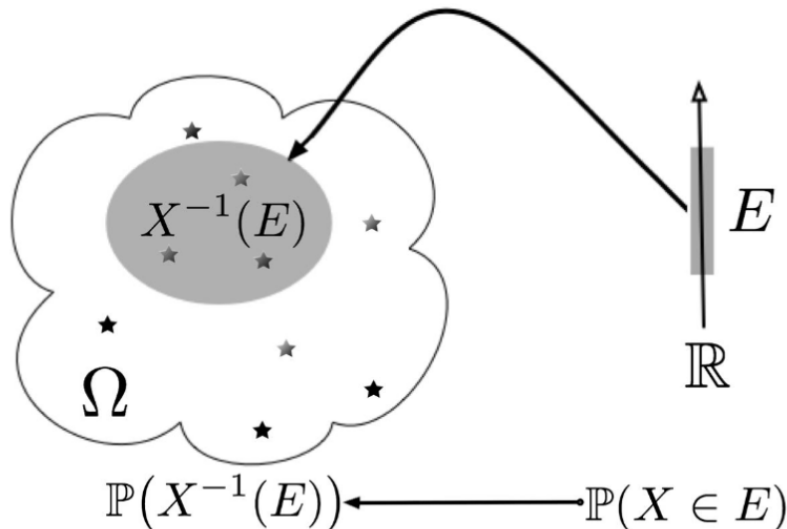
Here  $X$  is the random variable, while  $x$  is a particular real number.

#### Definition 3.5: Distribution of a random variable

The *distribution* of  $X$  is the rule that tells us with what probability  $X$  takes values in a given set  $E$ .

$$P_X(E) = P(X \in E).$$

The set  $E$  contains values;  $X^{-1}(E)$  contains the outcomes that give those values. In our finite examples, every subset of  $\Omega$  is an event.



A preimage and its probability. Illustration from Brutti's slides.

### Example 9: Which students have heights in an interval?

Choose one student uniformly from  $\Omega = \{\text{Anna, Luca, Sara, Marco}\}$ . Let  $X$  record height in centimetres:

| $\omega$    | Anna | Luca | Sara | Marco |
|-------------|------|------|------|-------|
| $X(\omega)$ | 165  | 178  | 178  | 190   |

For  $E = [170, 180]$ , Luca and Sara are the students whose heights fall inside the interval. Thus

$$X^{-1}([170, 180]) = \{\text{Luca, Sara}\}, \quad P(170 \leq X \leq 180) = \frac{2}{4} = \frac{1}{2}.$$

The same two students also give the exact value 178:  $X^{-1}(\{178\}) = \{\text{Luca, Sara}\}$ , so  $P(X = 178) = 1/2$ . One value can correspond to several outcomes.

### Example 10: Counting heads in two coin tosses

Toss a fair coin twice, with independent tosses. The four equally likely outcomes are  $\Omega = \{HH, HT, TH, TT\}$ , where  $H$  means heads and  $T$  means tails. Let  $X$  count the heads:

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

To get exactly one head, we need either  $HT$  or  $TH$ . Therefore

$$X^{-1}(\{1\}) = \{HT, TH\}, \quad P(X = 1) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}.$$

**Example 11: Counting tails in five coin tosses**

Toss a fair coin five times, with independent tosses. Write  $H$  for heads and  $T$  for tails. The sample space is  $\Omega = \{H, T\}^5$ , containing  $2^5 = 32$  sequences. Let  $X$  count the tails in a sequence, so its possible values are 0, 1, 2, 3, 4, 5.

Each particular sequence has probability  $(1/2)^5 = 1/32$ . For example,  $HHHHH$  and  $TTHHH$  are equally likely, although they give different values of  $X$ .

**No tails.** Only  $HHHHH$  gives zero tails:

$$\begin{aligned} P(X = 0) &= P(X^{-1}(\{0\})) \\ &= P(\{HHHHH\}) \\ &= \binom{5}{0} \left(\frac{1}{2}\right)^5 = \frac{1}{32}. \end{aligned}$$

**One tail.** The tail can occupy any of the five positions:

$$\begin{aligned} P(X = 1) &= P(X^{-1}(\{1\})) \\ &= P(\{THHHH, HTHHH, HHTHH, HHHTH, HHHHT\}) \\ &= \binom{5}{1} \left(\frac{1}{2}\right)^5 = \frac{5}{32}. \end{aligned}$$

**Two tails.** Choose which two of the five positions contain tails. The corresponding event contains  $\binom{5}{2} = 10$  sequences:

$$\begin{aligned} P(X = 2) &= P(X^{-1}(\{2\})) \\ &= P(\{TTHHH, THTHH, \dots, HHHTT\}) \\ &= \binom{5}{2} \left(\frac{1}{2}\right)^5 = \frac{10}{32}. \end{aligned}$$

In general, exactly  $x$  tails can occur in  $\binom{5}{x}$  sequences. Summing their probabilities gives

$$P(X = x) = P(X^{-1}(\{x\})) = \binom{5}{x} \left(\frac{1}{2}\right)^5, \quad x \in \{0, 1, 2, 3, 4, 5\},$$

The two factors are the number of sequences with  $x$  tails and the probability of each sequence.

**Example 12: Darts: distance from the centre**

Throw a dart at a circular target of radius  $r > 0$ . Assume that the dart always hits the disc and that its hitting point is uniform over the surface: regions with the same area have the same probability of being hit.

Place the centre at the origin. An outcome is the hitting point  $\omega = (u, v)$ , so the sample space is

$$\Omega = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 \leq r^2\}.$$

Let  $X$  measure the distance from that point to the centre:

$$X : \Omega \rightarrow \mathbb{R}, \quad X(u, v) = \sqrt{u^2 + v^2}.$$

For a target of radius 10 cm, some possible outcomes and their distances are:

| Hitting point $\omega$ | Distance $X(\omega)$ (cm) |
|------------------------|---------------------------|
| (0, 0)                 | 0                         |
| (3, 4)                 | $\sqrt{3^2 + 4^2} = 5$    |
| (-3, 4)                | $\sqrt{(-3)^2 + 4^2} = 5$ |
| (0, 10)                | 10                        |

In general, the possible values of  $X$  fill the interval  $[0, r]$ : zero at the centre,  $r$  on the boundary, and any distance in between.

For the 10 cm target, asking for a distance of at most 5 cm selects the outcomes

$$X^{-1}([0, 5]) = \{(u, v) \in \Omega : u^2 + v^2 \leq 25\}.$$

This is the smaller disc of radius 5 cm around the centre.

Now choose a distance threshold  $x$ . For  $0 \leq x \leq r$ , the event  $X \leq x$  means that the dart hits the inner disc of radius  $x$ . Since the hitting point is uniform over the target, its probability is the ratio of the two areas:

$$P(X \leq x) = \frac{\text{area of the inner disc}}{\text{area of the target}} = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2}.$$

A negative threshold is impossible for a distance, while a threshold larger than  $r$  includes the entire target. Thus

$$P(X \leq x) = \begin{cases} 0, & x < 0, \\ x^2/r^2, & 0 \leq x \leq r, \\ 1, & x > r. \end{cases}$$

For the target of radius 10 cm,

$$P(X \leq 5) = \frac{\pi \cdot 5^2}{\pi \cdot 10^2} = \frac{25}{100} = \frac{1}{4}.$$

## 3.4 Discrete random variables

### Definition 3.6: Discrete random variable and PMF

A random variable  $X$  is *discrete* if there is a finite or countably infinite set  $\mathcal{X} = \{x_1, x_2, \dots\}$  such that  $P(X \in \mathcal{X}) = 1$ .

Its *probability mass function* (PMF) is

$$p_X(x) := P(X = x), \quad x \in \mathbb{R}.$$

1.  $p_X(x) \geq 0$  for every  $x \in \mathbb{R}$ .
2. The *support*, in the discrete sense used here, is  $\text{supp}(X) := \{x \in \mathbb{R} : p_X(x) > 0\}$ . Thus the support lists the values that have positive probability; values outside it have probability zero.
3.  $\sum_{x \in \mathcal{X}} p_X(x) = 1$ .

By countable additivity, for any set  $E$ ,

$$P(X \in E) = \sum_{x \in E \cap \text{supp}(X)} p_X(x).$$

### Example 13: The PMF of a fair die

Roll a fair six-sided die and let  $X$  be the number shown. The random variable  $X$  is discrete because the finite set  $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$  satisfies  $P(X \in \mathcal{X}) = 1$ .

Its *probability mass function* (PMF) is:

$$p_X(x) = \begin{cases} 1/6, & x \in \{1, 2, 3, 4, 5, 6\}, \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies  $p_X(x) \geq 0$  for every  $x$ , and  $\sum_{x \in \mathcal{X}} p_X(x) = 6 \cdot (1/6) = 1$ .

Its *support* is  $\text{supp}(X) = \{1, 2, 3, 4, 5, 6\}$ , since each of these values has positive probability. For example,  $p_X(3) = 1/6$ , whereas  $p_X(7) = 0$  and 7 is outside the support.

Let's see an example that shows how to use the PMF to calculate the probability of an event. To find the probability of rolling an even number, we add the probabilities of rolling 2, 4, or 6:

$$P(X \in \{2, 4, 6\}) = p_X(2) + p_X(4) + p_X(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}.$$

---

\*A finite set has finitely many elements. A set  $A$  is countably infinite if there exists a bijection  $f : \mathbb{N} \rightarrow A$ .

## 3.5 Continuous random variables

### Definition 3.7: Continuous random variable and PDF

A random variable  $X$  is *continuous* (in the sense used here) if there exists a function  $f_X$ , called its *probability density function* (PDF), such that, for every  $a \leq b$ ,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

1.  $f_X(x) \geq 0$  for every  $x \in \mathbb{R}$ .
2. Following the convention in the slides, the *support* is  $\text{supp}(X) := \{x \in \mathbb{R} : f_X(x) > 0\}$ .
3.  $\int_{\mathbb{R}} f_X(x) dx = 1$ .

### Note: Heads-up

**A PDF cannot concentrate probability on a countable set.** First, setting  $a = b = x_i$  in the definition gives

$$P(X = x_i) = \int_{x_i}^{x_i} f_X(t) dt = 0 :$$

an interval of zero length has zero area under the PDF.

Now let  $A = \{x_1, x_2, \dots\}$  be any finite or countably infinite set, listing its elements without repetitions. The event  $X \in A$  is the union of the pairwise disjoint events  $\{X = x_i\}$ . By countable additivity,

$$P(X \in A) = \sum_i P(X = x_i) = \sum_i 0 = 0.$$

Thus no finite or countably infinite set can contain  $X$  with probability 1. The existence of a PDF already implies this; having an uncountable set of possible values alone does not guarantee that a PDF exists.

**Example 14: Waiting for a bus**

Suppose a bus arrives every 10 minutes and you reach the stop at a uniformly random time between two arrivals. Let  $X$  be your waiting time in minutes. Intervals of the same length have the same probability, so the PDF is constant on  $[0, 10]$ . Its height is  $1/10$ , since the total area must be 1.

Its PDF is:

$$f_X(x) = \begin{cases} 1/10, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies  $f_X(x) \geq 0$  for every  $x$ , and  $\int_{\mathbb{R}} f_X(x) dx = \int_0^{10} (1/10) dx = 1$ .

Its *support*, using the convention above, is  $[0, 10]$ .

Let's see an example that shows how to use the PDF to calculate the probability of an event. The probability of waiting between 2 and 5 minutes is:

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{10} dx = \frac{5-2}{10} = 0.30.$$

## 3.6 Cumulative distribution function (CDF)

### Definition 3.8: Cumulative distribution function

Given a random variable  $X$ , its *cumulative distribution function* (CDF) is the function  $F_X : \mathbb{R} \rightarrow [0, 1]$  defined by

$$F_X(x) := P(X \leq x) = P(X \in (-\infty, x]), \quad x \in \mathbb{R}.$$

It gives the probability accumulated up to and including the threshold  $x$ . It is defined for both [discrete](#) and [continuous](#) random variables.

### Proposition: The CDF determines the distribution

The CDF contains all the information about the *distribution* of a random variable. If  $F_X(t) = F_Y(t)$  for every  $t \in \mathbb{R}$ , then  $X$  and  $Y$  have the same distribution: for any set of values  $E$  defining an event,

$$P(X \in E) = P(Y \in E).$$

This does not mean that  $X$  and  $Y$  must produce the same outcome in an experiment.

*Proof.* We prove only a special case of this statement: we show that it holds for intervals  $E = (a, b]$ . The full result also holds for any set  $E$  defining an event, but we do not prove that general case here.

Suppose  $F_X(t) = F_Y(t)$  for every  $t \in \mathbb{R}$ , and let  $a < b$ . Since

$$\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\},$$

and the two events on the right are disjoint, additivity gives

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F_X(b) - F_X(a).$$

Applying the same reasoning to  $Y$  and using the equality of the two CDFs, we obtain

$$\begin{aligned} P(X \in E) &= F_X(b) - F_X(a) \\ &= F_Y(b) - F_Y(a) \\ &= P(Y \in E). \end{aligned}$$

□

### Note: PMF and CDF

The [PMF](#) answers: “What is the probability that  $X$  is exactly  $x$ ?”

$$p_X(x) = P(X = x).$$

The [CDF](#) answers: “What is the probability that  $X$  is at most  $x$ ?”

$$F_X(x) = P(X \leq x).$$

### 3.6.1 Properties of the CDF

#### Proposition 3.1: Properties of the CDF (I)

A function  $F(\cdot)$  mapping the real line to  $[0, 1]$  is a CDF for some probability measure  $P(\cdot)$  if and only if:

1.  $F(\cdot)$  is *non-decreasing*:  $x_1 < x_2$  implies that  $F(x_1) \leq F(x_2)$ .
2.  $F(\cdot)$  is *normalized*:

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1.$$

3.  $F(\cdot)$  is *right-continuous*: for all  $x$  we have

$$F(x) = F(x^+), \quad F(x^+) := \lim_{t \downarrow x} F(t).$$

#### Proposition 3.2: Properties of the CDF (II/a)

Let  $F_X(\cdot)$  be the CDF of a random variable  $X$ . Then:

1. For  $x_1 < x_2$ ,

$$P(x_1 < X \leq x_2) = P(X \in (x_1, x_2]) = F_X(x_2) - F_X(x_1).$$

In the discrete case, be careful with the strict and non-strict inequalities.

- 2.

$$P(X > x) = P(X \in (x, +\infty)) = 1 - F_X(x).$$

3. If  $X$  is continuous with PDF  $f_X(\cdot)$ , then

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

A PDF can be recovered from the CDF as

$$f_X(x) = F'_X(x)$$

at points where  $F_X$  is differentiable.

**Proposition 3.3: Properties of the CDF (II/b)**

Let  $F_X(\cdot)$  be the CDF of a random variable  $X$ . Then:

1. 
$$P(x_1 < X \leq x_2) = P(X \in (x_1, x_2]) = F_X(x_2) - F_X(x_1), \quad x_1 < x_2.$$
2. 
$$P(X > x) = P(X \in (x, +\infty)) = 1 - F_X(x).$$
3. If  $X$  is discrete with PMF  $p_X(\cdot)$  and support  $\mathcal{X} = \{x_1, x_2, \dots\}$ , then

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) = P(X < x) + p_X(x).$$

Hence

$$p_X(x) = F_X(x) - F_X(x^-), \quad F_X(x^-) := \lim_{t \uparrow x} F_X(t).$$

Let  $F_X(\cdot)$  be the CDF of a continuous random variable with PDF  $f_X(\cdot)$ . The following informal observations motivate later nonparametric statistical developments:

1. Roughly speaking,  $F_X(\cdot)$  is “simpler” than  $f_X(\cdot)$ .
2. It must be non-decreasing, so its shape is more constrained than that of a PDF.
3. Integration accumulates and smooths the PDF, making the CDF more regular.
4. This suggests that estimating the CDF is easier than estimating the PDF: fewer observations may be needed to reach the same statistical accuracy.

**Example 15: Five coin tosses: PMF and CDF**

Return to five independent flips of a balanced coin, and let  $X$  count the tails. Its possible values are 0, 1, 2, 3, 4, 5.

Its *probability mass function* (PMF) is:

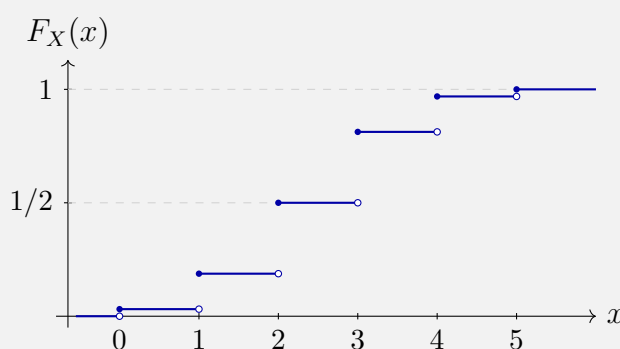
$$p_X(x) = P(X = x) = \binom{5}{x} \left(\frac{1}{2}\right)^5, \quad x \in \{0, 1, 2, 3, 4, 5\},$$

and  $p_X(x) = 0$  otherwise. The probabilities are:

| $x$      | 0    | 1    | 2     | 3     | 4    | 5    |
|----------|------|------|-------|-------|------|------|
| $p_X(x)$ | 1/32 | 5/32 | 10/32 | 10/32 | 5/32 | 1/32 |

Its *cumulative distribution function* (CDF) is obtained by adding the probabilities of all values at most  $x$ :

$$F_X(x) = \sum_{k \leq x} p_X(k) = \begin{cases} 0, & x < 0, \\ 1/32, & 0 \leq x < 1, \\ 6/32, & 1 \leq x < 2, \\ 16/32, & 2 \leq x < 3, \\ 26/32, & 3 \leq x < 4, \\ 31/32, & 4 \leq x < 5, \\ 1, & x \geq 5. \end{cases}$$



Filled points give the value of the CDF at each jump; open points are excluded.

The graph displays the three properties in [Properties of the CDF \(I\)](#):

1. *Non-decreasing*: each jump adds a positive probability.
2. *Normalized*:  $F_X(x) = 0$  for  $x < 0$  and  $F_X(x) = 1$  for  $x \geq 5$ .
3. *Right-continuous*: for example, consider  $x = 2$ . We have  $F_X(2) = 16/32 = 1/2$ , and the same value holds for  $2 \leq x < 3$ . Thus the right limit equals  $F_X(2)$ .

**Example 16: Darts: from areas to the CDF and PDF**

Return to the [earlier darts example](#), with a disc of radius 10 cm and a hitting point uniform over its area. Let  $X$  be the distance from the centre in centimetres. Earlier, we obtained  $P(X \leq x)$  using areas. Now we recognise this function as the CDF, differentiate it to find a PDF, and use that PDF to calculate probabilities of intervals.

**1. Recognise the CDF.** The function calculated earlier is the CDF:

$$F_X(x) = P(X \leq x) = \begin{cases} 0, & x < 0, \\ x^2/100, & 0 \leq x \leq 10, \\ 1, & x > 10. \end{cases}$$

**2. Derive the PDF.** The PDF is not given. We find it by differentiating the CDF for  $0 < x < 10$ :

$$f_X(x) = F'_X(x) = \frac{d}{dx} \left( \frac{x^2}{100} \right) = \frac{2x}{100} = \frac{x}{50}.$$

Its PDF is:

$$f_X(x) = \begin{cases} x/50, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

This PDF reproduces the probabilities of the model. It is non-negative and satisfies

$$\int_{\mathbb{R}} f_X(x) dx = \int_0^{10} \frac{x}{50} dx = \left[ \frac{x^2}{100} \right]_0^{10} = 1.$$

Its support, with the convention used above and this choice of PDF, is  $(0, 10]$ .

**3. Use the PDF to calculate an interval probability.** Now that we have derived the PDF, we can, for example, calculate the probability that the dart lands between 5 and 8 cm from the centre:

$$P(5 \leq X \leq 8) = \int_5^8 \frac{x}{50} dx = \left[ \frac{x^2}{100} \right]_5^8 = \frac{64 - 25}{100} = 0.39.$$

The CDF gives the same result:  $F_X(8) - F_X(5) = 0.64 - 0.25 = 0.39$ .

For the two intervals used in the slides, the same method gives

$$P(0 < X \leq 5) = \int_0^5 \frac{x}{50} dx = \frac{1}{4}, \quad P(5 < X \leq 10) = \int_5^{10} \frac{x}{50} dx = \frac{3}{4}.$$

**End of lecture 25 September**

Start of lecture 30 September

## 3.7 Relations

### 3.7.1 From the PMF to the CDF

#### What we are trying to answer

If we know the probability of each individual value of  $X$ , how do we calculate the probability that  $X$  is at most  $x$ ?

For a discrete random variable, the event  $X \leq x$  includes every possible value  $k$  that is at most  $x$ . These events  $\{X = k\}$  are pairwise disjoint, so we add their probabilities:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \sum_{\substack{k \in \text{supp}(X) \\ k \leq x}} \underbrace{p_X(k)}_{\text{PMF: known}}, \quad x \in \mathbb{R}.$$

### 3.7.2 From the PDF to the CDF

#### What we are trying to answer

We know how probability is spread over the possible values of  $X$  (its PDF). How do we calculate the probability of getting a value less than or equal to  $x$ ?

For a continuous random variable with PDF  $f_X$ , we calculate  $P(X \leq x)$  by integrating the PDF over  $(-\infty, x]$ :

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \int_{-\infty}^x \underbrace{f_X(t)}_{\text{PDF: known}} dt, \quad x \in \mathbb{R}.$$

### 3.7.3 From the CDF to the PMF and PDF

#### What we are trying to answer

We know the CDF  $F_X$ . How do we recover the PMF  $p_X$  if  $X$  is discrete, or the PDF  $f_X$  if  $X$  is continuous?

**What we know.** The CDF is defined by

$$F_X(x) := P(X \leq x).$$

For every threshold  $x$ , it gives the probability of obtaining a value less than or equal to  $x$ . We want to recover the PMF or the PDF, depending on the type of  $X$ .

**Discrete case: recover the PMF.** The PMF is defined by

$$p_X(x) := P(X = x).$$

The CDF includes all values up to  $x$ . To isolate the probability of the single value  $x$ , we subtract the probability of values strictly less than  $x$ :

$$P(X = x) = P(X \leq x) - P(X < x).$$

Since the left limit satisfies

$$F_X(x^-) := \lim_{t \uparrow x} F_X(t) = P(X < x),$$

we obtain

$$\underbrace{p_X(x)}_{\text{PMF: to find}} = \underbrace{F_X(x) - F_X(x^-)}_{\text{CDF: known}}.$$

Thus the PMF at  $x$  is the size of the jump of the CDF at  $x$ .

**Continuous case: recover the PDF.** A PDF is a non-negative function  $f_X$  such that

$$P(a \leq X \leq b) = \int_a^b f_X(t) dt, \quad \int_{\mathbb{R}} f_X(t) dt = 1.$$

In particular,

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt.$$

The CDF is therefore the integral of the PDF. To recover a PDF, we differentiate the CDF:

$$\underbrace{f_X(x)}_{\text{PDF: to find}} = \frac{d}{dx} \underbrace{F_X(x)}_{\text{CDF: known}}.$$

This recovers a PDF at points where the derivative exists, by the fundamental theorem of calculus.

## 3.8 The quantile function

### What we are trying to answer

Given a probability  $p$ , which threshold  $x$  satisfies  $P(X \leq x) = p$ ?

The CDF starts with a threshold and returns a probability. The quantile function starts with a probability and returns a threshold. For example, the 90th percentile is a threshold below which 90% of the probability lies; the median is the quantile at level 0.5.

### Definition 3.9: Quantile function: continuous case

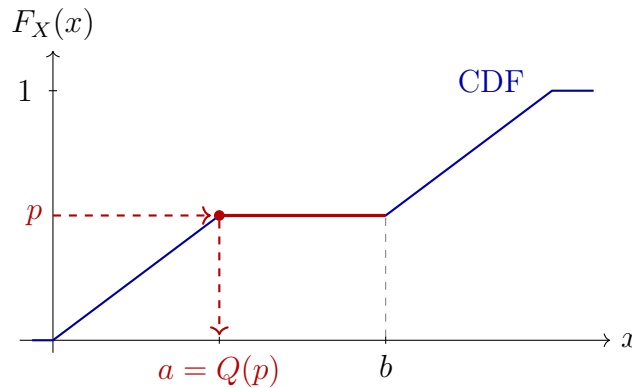
Let  $X$  be a continuous random variable with CDF  $F_X$ . For a probability level  $0 < p < 1$ , its *quantile function* is

$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) = p\}.$$

It gives the smallest value  $x$  such that  $F_X(x) = P(X \leq x) = p$ . We also write  $q_p$  instead of  $Q(p)$ .

The symbol  $\inf$  denotes the infimum (greatest lower bound); when the set has a minimum, the infimum is its smallest element. Here we consider levels strictly between 0 and 1; endpoint levels may require separate conventions.

**Geometric interpretation.** Start at height  $p$  on the vertical axis, move horizontally to the CDF, and then read the corresponding threshold on the horizontal axis. If several thresholds give the same probability, choose the smallest one.



In this continuous CDF, every threshold in  $[a, b]$  gives the same probability  $p$ . Thus  $\{x : F_X(x) = p\} = [a, b]$  and  $Q(p) = a$ .

When the CDF is strictly increasing between probability levels 0 and 1, each level corresponds to a unique threshold. The CDF is therefore invertible on this range, so

$$Q(p) = F_X^{-1}(p).$$

An inverse may exist without a simple explicit formula. In that case, quantiles can be calculated numerically.

**A more general definition.** If the CDF has a jump, it may skip the level  $p$ , so the equation  $F_X(x) = p$  may have no solution. We therefore replace equality with  $F_X(x) \geq p$ .

**Definition 3.10: Quantile function: general case**

Let  $X$  be any random variable, discrete or continuous, with CDF  $F_X$ . For a probability level  $0 < p < 1$ , its *quantile function* is

$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) \geq p\}.$$

It gives the smallest threshold  $x$  such that  $P(X \leq x)$  is at least  $p$ . We also write  $q_p$  instead of  $Q(p)$ .

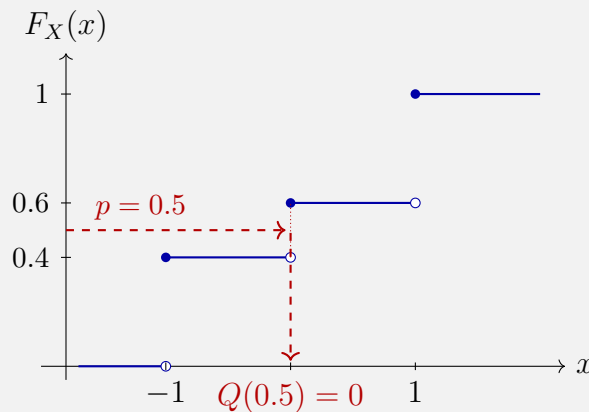
At a jump, the quantile is the first threshold where the CDF reaches or exceeds  $p$ ; the probability there may be greater than  $p$ . If the CDF has a plateau at level  $p$ , this definition chooses its left endpoint. This is the lower-quantile convention used here. If the CDF is invertible,  $Q(p) = F_X^{-1}(p)$  (see Definition 3.9).

**Example 17: Reading a quantile across a jump**

Let  $X$  take the values  $-1, 0, 1$  with probabilities  $0.4, 0.2, 0.4$ , respectively. Its CDF is

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 0.4, & -1 \leq x < 0, \\ 0.6, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

We want the median,  $Q(0.5)$ .



Start at height 0.5 and move to the right. Before  $x = 0$ , the CDF is below 0.5. At  $x = 0$ , the filled point gives  $F_X(0) = 0.6 \geq 0.5$ : this is the first threshold that qualifies. Therefore,

$$\{x : F_X(x) \geq 0.5\} = [0, +\infty), \quad \boxed{Q(0.5) = \inf[0, +\infty) = 0.}$$

Here  $P(X \leq Q(0.5)) = 0.6$ , which exceeds the requested level 0.5.

**Note: Quantiles as areas under the PDF**

**What we are trying to answer.** We know the PDF and choose a probability  $p$ . Where should we stop so that the area under the PDF to the left is  $p$ ?

For a continuous random variable with a PDF, we already know the [relation between the PDF and the CDF](#):

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt.$$

Previously, we fixed the threshold  $x$  and calculated the probability. Now we fix the probability  $p$  and look for the threshold  $q_p = Q(p)$ . Since the CDF is continuous,  $F_X(q_p) = p$ , so

$$\underbrace{p}_{\text{chosen probability}} = F_X(q_p) = \int_{-\infty}^{\overbrace{q_p}^{\text{threshold: to find}}} \underbrace{f_X(t)}_{\text{PDF: known}} dt.$$

Thus we find  $q_p$  by solving this equation. If several thresholds give the same area, the [general quantile definition](#) chooses the smallest one.

For the median,  $p = 0.5$ : we look for the threshold that leaves half of the total area under the PDF to its left.

**Example 18: Quantiles of a Pareto distribution**

The slides give a continuous random variable with CDF

$$F_X(x) = \begin{cases} 0, & x < 1, \\ 1 - x^{-\alpha}, & x \geq 1, \end{cases} \quad \alpha > 0.$$

This is a Pareto distribution with shape parameter  $\alpha$  and scale parameter 1. For example, if  $\alpha = 1/2$ , suppose we want to find the 20th percentile and the median.

**1. Recover the PDF.** For  $x > 1$ , we differentiate the given CDF:

$$f_X(x) = F'_X(x) = \alpha x^{-\alpha-1}.$$

We may therefore choose

$$f_X(x) = \begin{cases} \alpha/x^{\alpha+1}, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

**2. Find the quantile function.** The CDF is strictly increasing for  $x \geq 1$ . For  $0 < p < 1$ , we find the unique threshold  $q_p$  by solving  $F_X(q_p) = p$ :

$$\begin{aligned} p &= 1 - q_p^{-\alpha}, \\ q_p^{-\alpha} &= 1 - p, \\ q_p^\alpha &= \frac{1}{1 - p}, \\ q_p &= \left( \frac{1}{1 - p} \right)^{1/\alpha}. \end{aligned}$$

Thus the whole quantile function is

$$Q(p) = F_X^{-1}(p) = (1 - p)^{-1/\alpha}, \quad 0 < p < 1.$$

**3. Calculate the requested quantiles with  $\alpha = 1/2$ .** Here  $1/\alpha = 2$ , so  $Q(p) = (1 - p)^{-2}$ . For the 20th percentile, we choose  $p = 0.20$ :

$$q_{0.20} = (1 - 0.20)^{-2} = \frac{1}{0.8^2} = 1.5625 \approx 1.56.$$

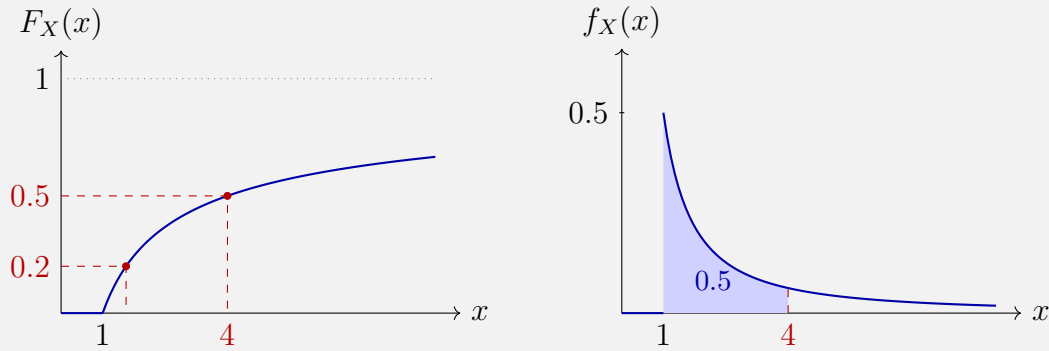
Hence  $P(X \leq 1.5625) = 0.20$ . For the median (the second quartile), we choose  $p = 0.50$ :

$$q_{0.50} = (1 - 0.50)^{-2} = \frac{1}{0.5^2} = 4.$$

Hence  $P(X \leq 4) = 0.50$ : the threshold 4 leaves half of the probability to its left.

**Note: Reading the Pareto quantiles geometrically**

For the Pareto example with  $\alpha = 1/2$ , the two graphs show the same median. On the CDF, height 0.5 corresponds to  $x = 4$ . Under the PDF, the area from 1 to 4 is 0.5.



The area calculation confirms the median:

$$\int_1^4 \frac{1}{2t^{3/2}} dt = \left[ -\frac{1}{\sqrt{t}} \right]_1^4 = -\frac{1}{2} + 1 = \frac{1}{2}.$$

**Example 19: Quantiles of a discrete distribution**

The second slide example uses the same variable as [the earlier geometric example](#):

|          |       |       |       |
|----------|-------|-------|-------|
| $x$      | $-1$  | $0$   | $1$   |
| $p_X(x)$ | $2/5$ | $1/5$ | $2/5$ |

We want the median, the 90th percentile, and the third quartile.

**1. Build the CDF by adding the masses.** At  $-1$  we have  $2/5$ ; at  $0$  we have  $2/5 + 1/5 = 3/5$ ; at  $1$  we have  $2/5 + 1/5 + 2/5 = 1$ . Therefore,

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 2/5 = 0.4, & -1 \leq x < 0, \\ 3/5 = 0.6, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

**2. Find the first threshold whose CDF reaches each level.** We use  $q_p = \inf\{x : F_X(x) \geq p\}$ .

- **Median:**  $p = 0.50$ . For  $x < 0$ ,  $F_X(x) \leq 0.4 < 0.5$ , whereas  $F_X(0) = 0.6 \geq 0.5$ . Thus

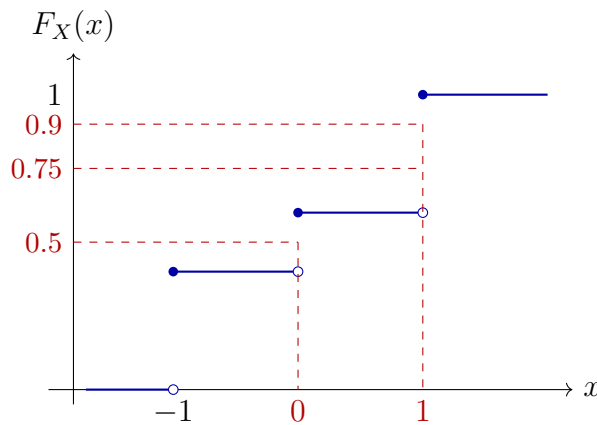
$$\{x : F_X(x) \geq 0.5\} = [0, +\infty), \quad \boxed{q_{0.50} = 0.}$$

- **90th percentile:**  $p = 0.90$ . For  $x < 1$ ,  $F_X(x) \leq 0.6 < 0.9$ , whereas  $F_X(1) = 1 \geq 0.9$ . Thus

$$\{x : F_X(x) \geq 0.9\} = [1, +\infty), \quad \boxed{q_{0.90} = 1.}$$

- **Third quartile:**  $p = 0.75$ . For  $x < 1$ ,  $F_X(x) \leq 0.6 < 0.75$ , whereas  $F_X(1) = 1 \geq 0.75$ . Thus

$$\{x : F_X(x) \geq 0.75\} = [1, +\infty), \quad \boxed{q_{0.75} = 1.}$$



The median is 0, while both the 90th percentile and the third quartile are 1: different probability levels can give the same quantile when the CDF has jumps.

## Chapter 3 (exercises)

### Exercise 4.1 — Two throws of a die

Source: Dekking et al., *A Modern Introduction to Probability and Statistics*, Section 4.6, p. 51.

Throw a fair die twice, independently. Let  $Z$  count the number of times a 6 appears. For an outcome  $\omega = (\omega_1, \omega_2)$ , define

$$S(\omega_1, \omega_2) = \omega_1 + \omega_2, \quad M(\omega_1, \omega_2) = \max\{\omega_1, \omega_2\}.$$

- (a) Describe the distribution of  $Z$  by giving either its PMF  $p_Z$  or its CDF  $F_Z$ . Identify the distribution family and its parameters.
- (b) List the outcomes in  $\{M = 2, Z = 0\}$ ,  $\{S = 5, Z = 1\}$ , and  $\{S = 8, Z = 1\}$ , and calculate their probabilities.
- (c) Determine whether  $\{M = 2\}$  and  $\{Z = 0\}$  are independent.

#### Part (a): the PMF of $Z$

Since  $Z$  counts the number of sixes in two throws,  $Z$  takes values in  $\{0, 1, 2\}$ . Formally, it is the random variable

$$Z : \{1, \dots, 6\}^2 \longrightarrow \mathbb{R}, \quad Z(\omega_1, \omega_2) = \mathbf{1}_{\{\omega_1=6\}} + \mathbf{1}_{\{\omega_2=6\}}.$$

Here each indicator is 1 if the corresponding throw is 6, and 0 otherwise.

To calculate the PMF, we proceed in two steps.

1. **First, show that  $Z$  is discrete.** Choose the finite set  $\mathcal{X} = \{0, 1, 2\}$ . Every outcome produces one of these values, so

$$P(Z \in \mathcal{X}) = P(\Omega) = 1.$$

Therefore,  $Z$  is discrete.

2. **Next, calculate its PMF.** By definition,  $p_Z(x) = P(Z = x)$  for  $x \in \mathbb{R}$ . There are 36 equally likely ordered outcomes.

If  $x = 0$ , neither throw is 6: each throw has 5 possible values.

$$P(Z = 0) = \frac{5 \cdot 5}{36} = \frac{25}{36}.$$

---

If  $x = 1$ , exactly one throw is 6: it can be the first or the second.

$$P(Z = 1) = \frac{1 \cdot 5 + 5 \cdot 1}{36} = \frac{10}{36}.$$

If  $x = 2$ , both throws are 6: the only outcome is  $(6, 6)$ .

$$P(Z = 2) = \frac{1}{36}.$$

Hence its probability mass function (PMF) is:

$$p_Z(x) = \begin{cases} 25/36, & x = 0, \\ 10/36, & x = 1, \\ 1/36, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

### Part (a): calculating the CDF

Alternatively, we can use the cumulative distribution function,

$$F_Z(x) := P(Z \leq x).$$

Here  $\{Z \leq x\} = \{\omega \in \Omega : Z(\omega) \leq x\}$ : we collect the outcomes for which the number of sixes is at most  $x$ .

Since  $Z$  takes only the values 0, 1, 2, the included values change only when  $x$  reaches one of these numbers. We therefore distinguish  $x < 0$ ,  $0 \leq x < 1$ ,  $1 \leq x < 2$ , and  $x \geq 2$ .

For  $x < 0$ , no possible value of  $Z$  is included, so

$$F_Z(x) = P(Z \leq x) = P(\emptyset) = 0.$$

For  $0 \leq x < 1$ , only  $Z = 0$  is included:

$$F_Z(x) = P(Z \leq x) = P(Z = 0) = \frac{25}{36}.$$

In particular,  $F_Z(0) = 25/36$ .

For  $1 \leq x < 2$ , the included values are  $Z = 0$  and  $Z = 1$ :

$$\{Z \leq x\} = \{Z = 0\} \cup \{Z = 1\}.$$

These events are disjoint:  $Z$  cannot equal both 0 and 1 in the same outcome. By additivity of probability for disjoint events,

$$F_Z(x) = P(Z \leq x) = P(Z = 0) + P(Z = 1) = \frac{25}{36} + \frac{10}{36} = \frac{35}{36}.$$

In particular,  $F_Z(1) = 35/36$ .

---

For  $x \geq 2$ , all three possible values are included:

$$\{Z \leq x\} = \{Z = 0\} \cup \{Z = 1\} \cup \{Z = 2\} = \Omega.$$

These three events are pairwise disjoint. Therefore,

$$\begin{aligned} F_Z(x) = P(Z \leq x) &= P(Z = 0) + P(Z = 1) + P(Z = 2) \\ &= \frac{25}{36} + \frac{10}{36} + \frac{1}{36} = \frac{36}{36} = 1, \quad x \geq 2. \end{aligned}$$

Putting the cases together, its CDF is:

$$F_Z(x) = \begin{cases} 0, & x < 0, \\ 25/36, & 0 \leq x < 1, \\ 35/36, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

### Part (b): outcomes and probabilities

Both conditions in each event must hold for the same ordered pair. Each of the 36 ordered outcomes has probability  $1/36$ .

1.  $\{M = 2, Z = 0\}$ . If the maximum is 2, both throws are at most 2 and at least one is 2. All these outcomes also give  $Z = 0$ :

$$\{M = 2, Z = 0\} = \{(1, 2), (2, 1), (2, 2)\}.$$

Therefore,

$$P(M = 2, Z = 0) = \frac{3}{36} = \frac{1}{12}.$$

2.  $\{S = 5, Z = 1\}$ . Exactly one throw must be 6. The other is at least 1, so the sum is at least 7 and cannot be 5:

$$\{S = 5, Z = 1\} = \emptyset, \quad P(S = 5, Z = 1) = 0.$$

3.  $\{S = 8, Z = 1\}$ . One throw is 6 and the other must be 2:

$$\{S = 8, Z = 1\} = \{(6, 2), (2, 6)\}.$$

Therefore,

$$P(S = 8, Z = 1) = \frac{2}{36} = \frac{1}{18}.$$

---

**Part (c): independence**

Let  $A = \{M = 2\}$  and  $B = \{Z = 0\}$ . By the [definition of independence](#), we must check whether  $P(A \cap B) = P(A)P(B)$ .

If  $M = 2$ , neither throw can be 6, so  $Z = 0$ . Thus  $A \subseteq B$  and  $A \cap B = A$ . Using the counts above,

$$P(A \cap B) = P(A) = \frac{3}{36} = \frac{1}{12}.$$

On the other hand,

$$P(A)P(B) = \frac{3}{36} \cdot \frac{25}{36} = \frac{25}{432} \neq \frac{1}{12}.$$

Therefore,  $\{M = 2\}$  and  $\{Z = 0\}$  are **not independent**. Equivalently, knowing that  $M = 2$  guarantees that  $Z = 0$ :

$$P(Z = 0 \mid M = 2) = 1 \neq \frac{25}{36} = P(Z = 0).$$

# 4

## A tour of well-known distributions

### 4.1 The uniform distribution

A **uniform distribution** assigns the same probability to intervals of the same length within its support. Its parameters  $a$  and  $b$  specify the ends of that support.

#### Definition 4.1: Uniform distribution

A continuous random variable  $X$  has a *uniform distribution* with parameters  $a < b$ , where  $a, b \in \mathbb{R}$ , if its PDF and CDF are

$$f_X(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x),$$
$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

We denote this distribution by  $\text{Unif}(a, b)$  and write  $X \sim \text{Unif}(a, b)$ . Here  $\text{Unif}$  abbreviates *uniform*, and  $\sim$  means *has distribution*.

**Why is the density a rectangle?** Uniformity means that intervals of the same length within  $[a, b]$  have the same probability, hence the same area under the PDF. If the density were higher in one region, a small interval there would have more probability than an equally long interval in a lower region. The density is therefore constant on  $[a, b]$  and zero outside it. Its graph is horizontal on the support, so the region beneath it is a rectangle: the rectangular shape follows from uniformity, before we calculate its height.

**Why is the height  $1/(b-a)$ ?** Let  $h$  be the constant height. The rectangle has width  $b-a$ , and the total area under a PDF must be 1:

$$h(b-a) = 1 \quad \implies \quad h = \frac{1}{b-a}.$$

For  $a \leq c \leq d \leq b$ ,

$$P(c \leq X \leq d) = \int_c^d \frac{1}{b-a} dx = \frac{d-c}{b-a}.$$

The numerator is the length of the selected interval; the denominator is the length of the full support.

For  $a \leq x \leq b$ , the same calculation gives the CDF:

$$F_X(x) = \int_a^x \frac{1}{b-a} dt = \frac{x-a}{b-a}.$$

#### 4.1.1 The standard uniform distribution: $\text{Unif}(0, 1)$

The **standard uniform distribution**, written  $U \sim \text{Unif}(0, 1)$ , is the **special case** of the general  $\text{Unif}(a, b)$  distribution in Section 4.1 obtained by setting  $a = 0$  and  $b = 1$ .

Its support is therefore  $[0, 1]$ , and substituting these endpoints gives the constant density  $1/(b-a) = 1/(1-0) = 1$ . Its PDF is

$$f_U(u) = \begin{cases} 1, & 0 \leq u \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

For  $0 \leq c \leq d \leq 1$ ,

$$P(c \leq U \leq d) = \int_c^d 1 du = d - c.$$

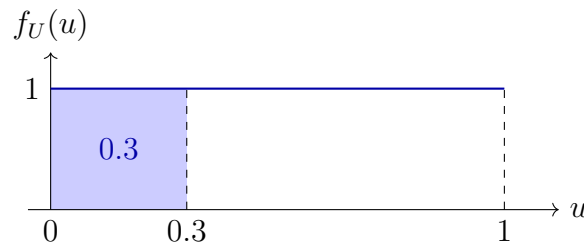
**Find the CDF.** By the [definition of a CDF](#),  $F_U(t) = P(U \leq t)$ . For  $0 \leq t \leq 1$ , integrate the PDF up to  $t$ :

$$F_U(t) = P(U \leq t) = \int_0^t 1 du = t.$$

This also follows directly from the general CDF formula:  $(t-a)/(b-a) = (t-0)/(1-0) = t$  for  $0 \leq t \leq 1$ . The full CDF is therefore

$$F_U(t) = \begin{cases} 0, & t < 0, \\ t, & 0 \leq t \leq 1, \\ 1, & t > 1. \end{cases}$$

For example,  $P(U \leq 0.3) = 0.3$ : the interval from 0 to 0.3 contains 30% of the probability. Geometrically, this is a rectangle with width 0.3 and height 1, so its area is 0.3.



# 5

## Simulation

Start of lecture 1 October

### 5.1 A “what if?” experiment

A “what if?” question asks what would happen in a hypothetical situation. A simulation reproduces possible outcomes of an experiment on a computer. We can repeat the simulation and observe what happens across the different repetitions.

#### Example 20: One hundred coin tosses

Suppose we ask:

If I tossed a coin 100 times, how many heads would I get?

A simulation produces a possible sequence of 100 tosses, and we count the heads in that sequence. We can repeat this experiment on the computer and compare the numbers of heads obtained.

### 5.2 From a real coin to a simulation

Imagine that we have a coin, but we do not know its probability of landing heads. For each toss, let  $X = 1$  mean heads and  $X = 0$  mean tails. Its distribution is described by

$$P(X = 1) = p, \quad P(X = 0) = 1 - p,$$

where  $p$  is unknown. This is our **population model**: it describes the possible outcomes and their probabilities.

We decide to toss the coin 100 times. We assume that the tosses are independent and that each has the same probability  $p$  of heads. This is our **sampling scheme**. We call the tosses **IID**: independent and identically distributed. The outcome of one toss gives no information about the others, and all tosses follow the same distribution.

Together, the population model and the sampling scheme describe the **data generating process**: the mechanism that produces the sequence of outcomes we observe.

Suppose we observe 65 heads in our 100 tosses. We use the observed proportion of heads to estimate the unknown probability:

$$\hat{p} = \frac{65}{100} = 0.65.$$

The hat means that  $\hat{p}$  is an estimate calculated from the data. It does not mean that the true  $p$  is exactly 0.65; another set of 100 tosses could give a different proportion. Using data to learn about an unknown model is called **inference**.

Now imagine that we choose the probability ourselves, for example  $p = 0.5$ , and ask the computer to produce 100 independent tosses with that probability. This is **simulation**: we specify the model and the sampling scheme, then generate data following those rules. We know the rules, but we do not fix the outcomes. One run could give 48 heads and another 53.

#### Note: Inference and simulation

**Inference.** We observe data and use them to learn about an unknown model.

Observed data  $\longrightarrow$  Information about the model.

**Simulation.** We choose a model and a sampling scheme, then generate data from them.

Specified model and sampling scheme  $\longrightarrow$  Simulated data.

For our coin, the simulation question is: what numbers of heads could we observe in 100 independent tosses if  $p = 0.5$ ? Repeating the experiment on the computer lets us explore the variation in those numbers.

## 5.3 Where do the random numbers come from?

We want the computer to simulate our 100 coin tosses. A useful starting point is a sequence of numbers that behave like independent draws from a uniform distribution on  $(0, 1)$ . But where do these numbers come from?

### 5.3.1 Using a physical source

A **true random number generator** measures a physical source of randomness, such as electronic noise or a quantum phenomenon. The device processes those measurements to

obtain numbers that a computer can use. The changing patterns of lava lamps are another example of a physical source.

We can distinguish two meanings of randomness:

- **Epistemic randomness:** we cannot predict an outcome because we lack information about the process.
- **Ontological randomness:** randomness is intrinsic to the physical phenomenon itself.

For simulation, the practical idea is to obtain numbers from a physical process. This requires a physical source and a device to read it. Collecting and processing measurements can take time. Repeating the experiment does not reproduce the same sequence of outcomes, although we can save the numbers already collected.

### 5.3.2 Using a formula

There is another approach: choose an initial value  $x_0$  and a function  $f$  that calculates the next value from the current one. Apply the same function repeatedly. This is how a **pseudorandom number generator** works.

$$\underbrace{x_0}_{\text{initial value}} \longrightarrow \underbrace{f(x_0)}_{x_1} \longrightarrow \underbrace{f(f(x_0))}_{x_2} \longrightarrow \underbrace{f(f(f(x_0)))}_{x_3} \longrightarrow \dots$$

First we calculate  $x_1 = f(x_0)$ . We then use  $x_1$  as the input to the same function, obtaining  $x_2 = f(x_1) = f(f(x_0))$ , and continue in this way. In general,  $x_{k+1} = f(x_k)$ : each output becomes the input for the next step. Here  $f$  denotes the update rule.

The procedure is **deterministic**: once the starting value and the rule are fixed, the entire sequence is determined. A suitable generator is designed so that the sequence imitates the statistical behaviour of random draws from the distribution we need.

The starting value is called the **seed**. Such generators can be fast and require no additional physical device. To reproduce a run, we can keep the seed and the generator settings rather than store the whole sequence.

#### Note: Why the seed matters

With the same generator, settings and seed, we obtain the same sequence. This makes a simulation **reproducible**.

Imagine that a simulation gives an unexpected result. We can run it again with the same seed and inspect exactly the same simulated experiment. We can also compare two programs using the same inputs.

These pseudorandom numbers are the starting material from which we generate simulated coin tosses, die rolls and observations from other distributions.

## 5.4 What should a pseudorandom sequence imitate?

We can now make the goal of the generator more precise. We want its output to imitate independent draws from a uniform distribution on  $(0, 1)$ .

### Definition 5.1: Pseudorandom number generator

In a simplified description, a *pseudorandom number generator* (PRNG) uses a deterministic function

$$f : (0, 1) \longrightarrow (0, 1)$$

and a starting value, or *seed*,  $u_0 \in (0, 1)$  to generate

$$u_0, \quad u_1 = f(u_0), \quad u_2 = f(f(u_0)), \quad \dots, \quad u_n = f^n(u_0).$$

The function is chosen so that the sequence imitates the statistical behaviour of an IID Uniform(0, 1) sample. Here  $f^n(u_0)$  means applying  $f$  repeatedly  $n$  times.

There are two parts to this goal:

- **Uniformity.** Intervals of the same length should receive approximately the same proportion of values. For example, in a sufficiently long sequence, about 10% of the values should fall in  $(0.2, 0.3)$ , and about 10% in  $(0.7, 0.8)$ .
- **Imitating independence.** The sequence should imitate draws in which observing one value gives no information about the others. We want to avoid systematic patterns between successive values.

For example, consider the repeating sequence

$$0.1, 0.2, 0.3, \dots, 0.9, 0.1, 0.2, \dots$$

It visits different parts of the interval, but its next value is easy to predict. Spreading values across the interval is only one part of what we require from a good generator.

### Note: The apparent paradox

We know the seed and the rule, so the whole sequence is determined. Yet we design the generator so that its output statistically resembles independent uniform draws. Imitating an IID sample is the goal; for a fixed seed, the generated sequence remains deterministic.

## 5.5 The linear congruential generator

A concrete way to generate the next value is to multiply the current value, add a constant, and keep the remainder after division.

**Definition 5.2: Linear congruential generator**

A *linear congruential generator* (LCG) uses integer parameters  $\alpha$ ,  $\beta$ , and  $M > 1$ , and an integer seed  $x_0 \in \{0, 1, \dots, M - 1\}$ . Its update rule is

$$x_{k+1} = f(x_k) = (\alpha + \beta x_k) \bmod M.$$

Here  $\bmod M$  gives the remainder after division by  $M$ , so every state  $x_k$  belongs to  $\{0, 1, \dots, M - 1\}$ .

At each step, we multiply the current value by  $\beta$ , add  $\alpha$ , and take the remainder. The resulting integer becomes the input for the next step.

**Example 21: Following the sequence and finding its period**

Choose small parameters so that we can follow the calculations:

$$\alpha = 1, \quad \beta = 5, \quad M = 16, \quad x_0 = 2.$$

The first three updates are

$$\begin{aligned} x_1 &= (1 + 5 \cdot 2) \bmod 16 = 11, \\ x_2 &= (1 + 5 \cdot 11) \bmod 16 = 56 \bmod 16 = 8, \\ x_3 &= (1 + 5 \cdot 8) \bmod 16 = 41 \bmod 16 = 9. \end{aligned}$$

For example,  $56 = 3 \cdot 16 + 8$ , so its remainder is 8. Continuing with the same rule gives

$$\begin{aligned} 2 &\rightarrow 11 \rightarrow 8 \rightarrow 9 \rightarrow 14 \rightarrow 7 \rightarrow 4 \rightarrow 5 \\ &\rightarrow 10 \rightarrow 3 \rightarrow 0 \rightarrow 1 \rightarrow 6 \rightarrow 15 \rightarrow 12 \rightarrow 13 \rightarrow 2. \end{aligned}$$

Once 2 returns, the same rule produces 11 again, then 8, and so on: the sequence repeats. The **period** is the length of this repeating cycle.

Here we visit all 16 possible states before returning to the starting value. The period is therefore 16, the maximum possible for  $M = 16$ . In general, an LCG can have period  $M$  with appropriate choices of its parameters.

To rescale the integer states to numbers between 0 and 1, we can use

$$u_k = \frac{x_k}{M + 1}.$$

For the first three generated states in the example, this gives

$$\frac{11}{17} \approx 0.647, \quad \frac{8}{17} \approx 0.471, \quad \frac{9}{17} \approx 0.529.$$

The output can be 0 when the remainder is 0, and is always less than 1. Rescaling changes the range of the values; the parameters determine how the sequence behaves statistically.

**Note: Why the choice of parameters matters**

We want the sequence to avoid repeating too soon. If we instead choose  $\beta = 2$ , keeping  $\alpha = 1$ ,  $M = 16$  and  $x_0 = 2$ , we get

$$2 \rightarrow 5 \rightarrow 11 \rightarrow 7 \rightarrow 15 \rightarrow 15 \rightarrow 15 \rightarrow \dots$$

The generator gets stuck at 15, so the repeating cycle has period 1. Such a sequence would be unsuitable for simulating coin tosses.

A long period is one requirement for a useful generator. We also need its output to imitate uniformity and independence. Our small example illustrates the calculations and a full cycle; assessing statistical quality requires more than checking the period.

## 5.6 The LCG in R

We can translate the update rule into a short R function. To follow the calculations, we keep the small parameters from [the previous example](#) and generate three values:

```
lcg <- function(n = 3, a = 1, b = 5, M = 16, u0 = 2) {
  # Map an integer seed to a valid state between 0 and M - 1.
  u0 <- u0 %% M
  out <- rep(NA, n)

  for (i in 1:n) {
    u0 <- (a + b*u0) %% M
    out[i] <- u0 / (M + 1)
  }

  return(out)
}
```

Here **a** corresponds to  $\alpha$ , **b** to  $\beta$ , and **n** is the number of values we ask the function to produce. The argument **u0** supplies the initial seed.

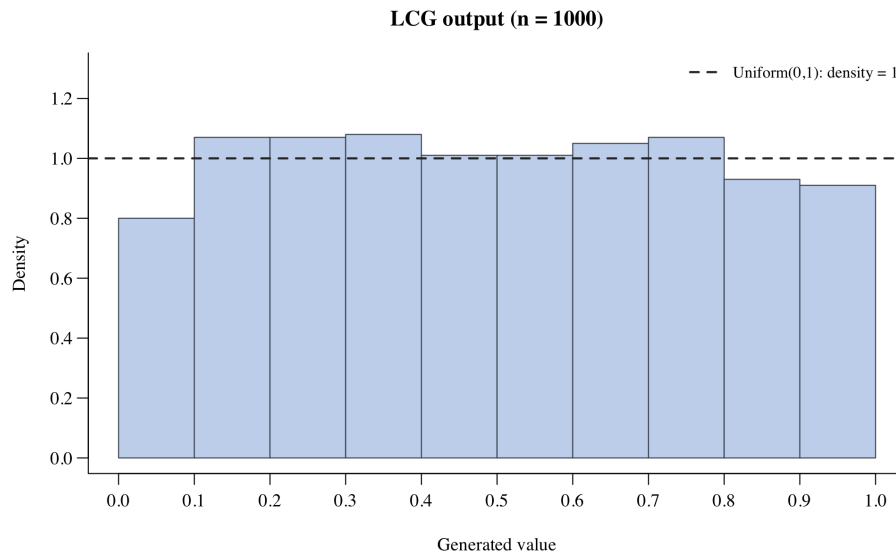
### 5.6.1 Checking uniformity with a histogram

A **histogram** shows how numerical values are spread across intervals. We divide the range into intervals, count the values in each one, and draw adjacent bars. For equal widths, taller bars mean more values: we can see where the values are concentrated.

We use the same function to generate 1000 values, with seed 26 and parameters  $a = 69069069$ ,  $b = 12345$  and  $M = 4096$ :

```
u <- lcg(n = 1000, a = 69069069, b = 12345,
        M = 2^12, u0 = 26)
hist(u, breaks = seq(0, 1, 0.1), probability = TRUE)
abline(h = 1, lty = 2)
```

We use ten intervals of width 0.1. Here 80 values fall between 0 and 0.1, and 107 between 0.1 and 0.2, so the first bar is lower than the second. For a uniform distribution on  $(0, 1)$ , each interval has probability 0.1: we would expect about 100 values per interval in a sample of size 1000.



With `probability = TRUE`, the vertical axis shows density. Each bar's height is its count divided by the product of the sample size and the interval width. For 100 values in an interval of width 0.1, the height is  $100/(1000 \times 0.1) = 1$ . Each bar's area is the fraction of values in its interval; the total area is 1.

The dashed line is the uniform density, 1 on  $(0, 1)$ . The bars are roughly close to it: a first visual check of uniformity. Their heights need not be identical in a finite sequence. A histogram does not check independence: rearranging the values leaves it unchanged.

## 5.7 Assessing the quality of a generator

We have seen how a generator produces numbers. We now ask whether those numbers behave well enough for our simulations.

Software uses generators more sophisticated than the LCG; the **Mersenne Twister** is one example. No pseudorandom generator produces a perfect IID sample: its sequence is determined by its initial state and update rule. Nevertheless, suitable generators can imitate independent uniform draws well enough for many practical purposes.

We assess three **properties** of the output, understood as criteria for judging the quality of the generator:

- **Property 0: Period.** The sequence should have a period long enough for the simulation. We examined this in [the LCG example](#).
- **Property 1: Distribution of samples.** The values should imitate draws from a uniform distribution on  $(0, 1)$ . The histogram provides a first visual check.
- **Property 2: Independence of samples.** Groups of values should imitate independent draws, without systematic patterns between successive outputs.

Producing numbers between 0 and 1 is only the starting point; these properties concern the behaviour of the sequence as a whole.

**Note: For further reading**

To prove when the LCG  $x_{k+1} = (\alpha + \beta x_k) \bmod M$  has period  $M$ , we need to establish four points:

1. **First necessary condition.** Show that period  $M$  requires  $\gcd(\alpha, M) = 1$ .
2. **Second necessary condition.** Show that period  $M$  requires  $\beta - 1$  to be divisible by every prime factor of  $M$ .
3. **Third necessary condition.** Show that, if 4 divides  $M$ , period  $M$  requires 4 to divide  $\beta - 1$ .
4. **Sufficiency.** Show that the three conditions together guarantee period  $M$  for every seed  $x_0 \in \{0, 1, \dots, M - 1\}$ .

Original reference: [Hull and Dobell, \*Random Number Generators\* \(1962\)](#).

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## 5.8 The quantile transform method

We now have a way to generate numbers that imitate independent uniform draws. How can we use them to simulate a different distribution? The link is the [quantile function](#).

Previously, we chose a probability level  $p$  and found the corresponding threshold  $Q(p)$ . Now we draw that level at random: we generate  $U$  uniformly between 0 and 1 and return  $Q(U)$ . This is the **quantile transform method**, also called the **inverse transform method**.

### Proposition 5.1: Quantile transform

Let  $F$  be any CDF and let

$$Q(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}, \quad 0 < p < 1,$$

be its quantile function. If  $U \sim \text{Unif}(0, 1)$  and  $Y = Q(U)$ , then  $Y$  has CDF  $F$ :

$$F_Y(y) = P(Y \leq y) = F(y), \quad y \in \mathbb{R}.$$

**Proof in the continuous, strictly increasing case.** In this case  $Q = F^{-1}$ . Since  $F$  is increasing,  $F^{-1}(U) \leq y$  is equivalent to  $U \leq F(y)$ . Therefore,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(F^{-1}(U) \leq y) \\ &= P(U \leq F(y)) \\ &= F(y). \end{aligned}$$

The last step uses the [standard uniform distribution's CDF](#):  $P(U \leq t) = t$  for  $0 \leq t \leq 1$ . Here the threshold is  $t = F(y)$ , which lies in  $[0, 1]$  because  $F$  is a CDF. Thus the generated variable has exactly the CDF we wanted. The result also holds for CDFs with jumps or flat parts, using the general quantile definition; the proof above covers the invertible case.

To generate a sample, draw independent  $U_1, \dots, U_n$  and set  $Y_i = Q(U_i)$  for each  $i$ . Applying the same function separately to independent draws preserves independence, and every  $Y_i$  has CDF  $F$ .

### 5.8.1 A continuous example

#### Example 22: Generating values with density $3x^2$

Suppose we want to simulate a variable  $X$  with PDF

$$f_X(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

**Find the CDF.** For  $0 \leq x < 1$ ,

$$F_X(x) = \int_0^x 3t^2 dt = [t^3]_0^x = x^3.$$

The complete CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^3, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

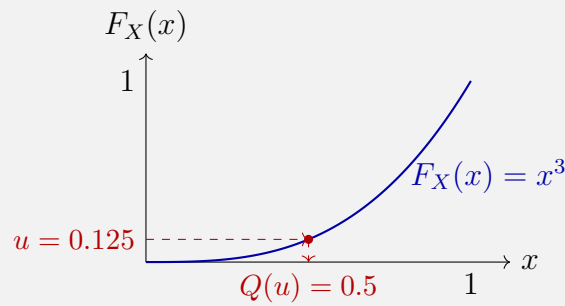
**Find the quantile.** For  $0 < p < 1$ , solve  $x^3 = p$ :

$$Q(p) = p^{1/3}.$$

**Generate a value.** Draw  $U \sim \text{Unif}(0, 1)$  and return

$$Y = U^{1/3}.$$

For example, the uniform value  $u = 0.125$  becomes  $y = 0.125^{1/3} = 0.5$ .



Geometrically, start at the randomly chosen height  $u$ , move right to the CDF, and read the corresponding value on the horizontal axis.

As a check, for  $0 \leq y \leq 1$ ,

$$P(Y \leq y) = P(U^{1/3} \leq y) = P(U \leq y^3) = y^3 = F_X(y).$$

## 5.8.2 A discrete example

### Example 23: Generating a Bernoulli variable

Suppose we want  $X \sim \text{Ber}(0.4)$ , so

$$P(X = 0) = 0.6, \quad P(X = 1) = 0.4.$$

Its CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.6, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

**Find the quantile.** This CDF has jumps. We use the smallest threshold where the CDF reaches or exceeds the chosen probability level:

$$Q(p) = \begin{cases} 0, & 0 < p \leq 0.6, \\ 1, & 0.6 < p < 1. \end{cases}$$

**Generate a value.** Draw  $U \sim \text{Unif}(0, 1)$  and return

$$Y = Q(U) = \begin{cases} 0, & U \leq 0.6, \\ 1, & U > 0.6. \end{cases}$$

The interval for output 0 has length 0.6; the interval for output 1 has length 0.4.



For example,  $u = 0.25$  gives  $y = 0$ , whereas  $u = 0.8$  gives  $y = 1$ . At  $u = 0.6$ , the general quantile definition gives  $y = 0$ .

**Check the probabilities.** Since  $U$  is uniform,

$$P(Y = 0) = P(U \leq 0.6) = 0.6, \quad P(Y = 1) = P(U > 0.6) = 1 - 0.6 = 0.4.$$

Thus  $Y$  has the same distribution as  $X$ . Equivalently,  $Y = \mathbf{1}_{(0.6, 1)}(U)$ : the indicator is 1 when  $U$  falls above 0.6, and 0 otherwise.